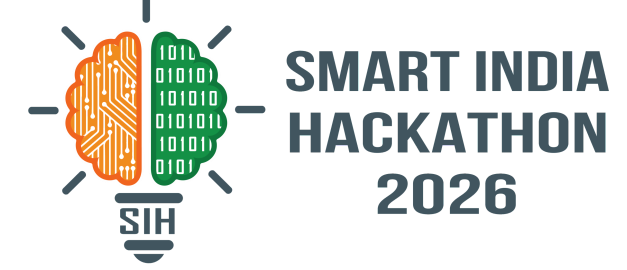


SMART INDIA HACKATHON 2026



- Problem Statement ID – **26141**
- Problem Statement Title- **Quantum-Inspired Cyber Threat Detection for Digital Signature Security**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID- **179587**
- Team Name - **Ghost_Protocol_**



SECURITY GAP

- **Quantum Threat:** RSA and ECC rest on computational hardness that Shor's algorithm dismantles.
- **Harvest-Now-Decrypt-Later:** adversaries already archive signed traffic to break retroactively.
- **Detection Blind Spot:** forgery, impersonation and replay all present as one undifferentiated failure.
- **Physical-Layer Exposure:** multi-photon pulses permit Photon Number Splitting, leaving no naive trace.

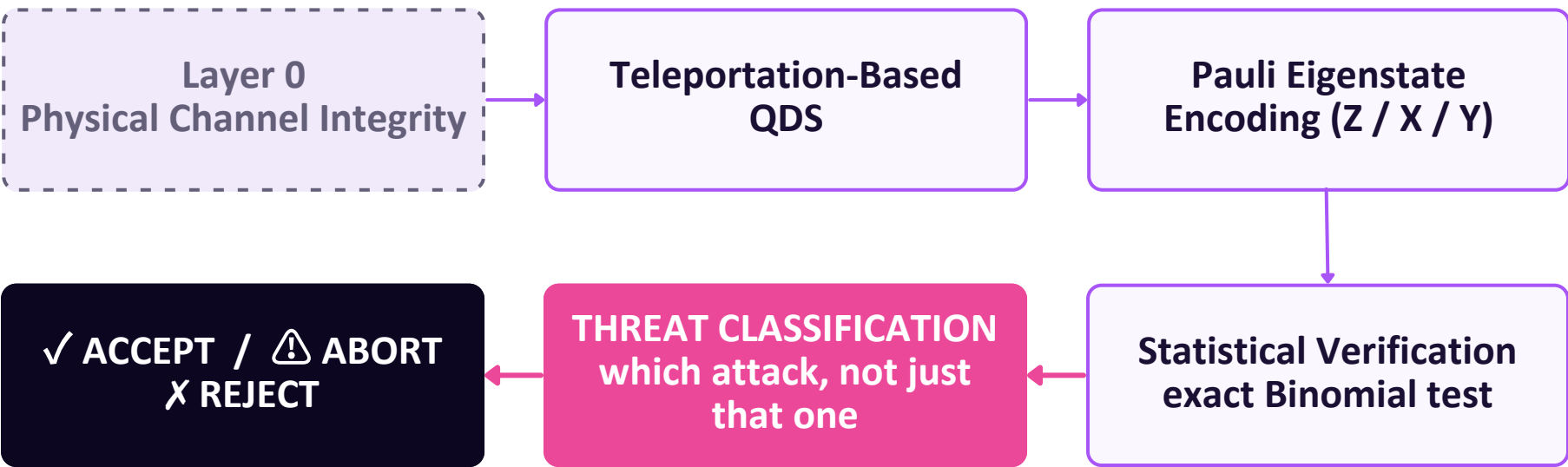
HOW IT ADDRESSES THE PROBLEM

- **Disturbance as a sensor:** no-cloning makes interception physically self-reporting.
- **Multi-basis verification:** Z/X/Y schedule makes basis-selective attacks identifiable, not just visible.
- **Exact statistics:** Binomial hypothesis testing against a calibrated noise baseline.
- **Three-way decision:** ACCEPT / ABORT / REJECT — an honest inconclusive band, never a forced guess.
- **Freshness binding:** a session nonce closes the same-message replay hole measurement cannot see.

THREAT CLASSES DETECTED AND IDENTIFIED



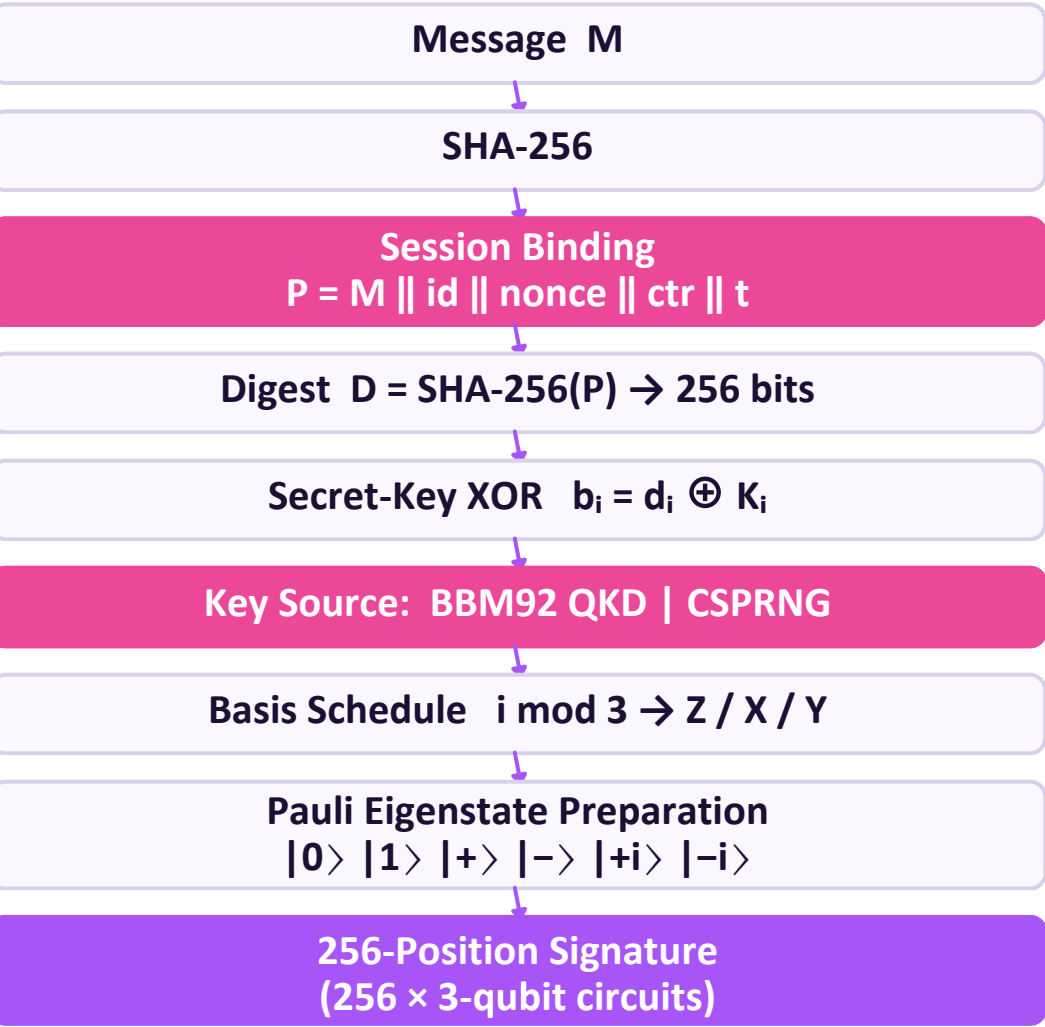
PROPOSED SOLUTION



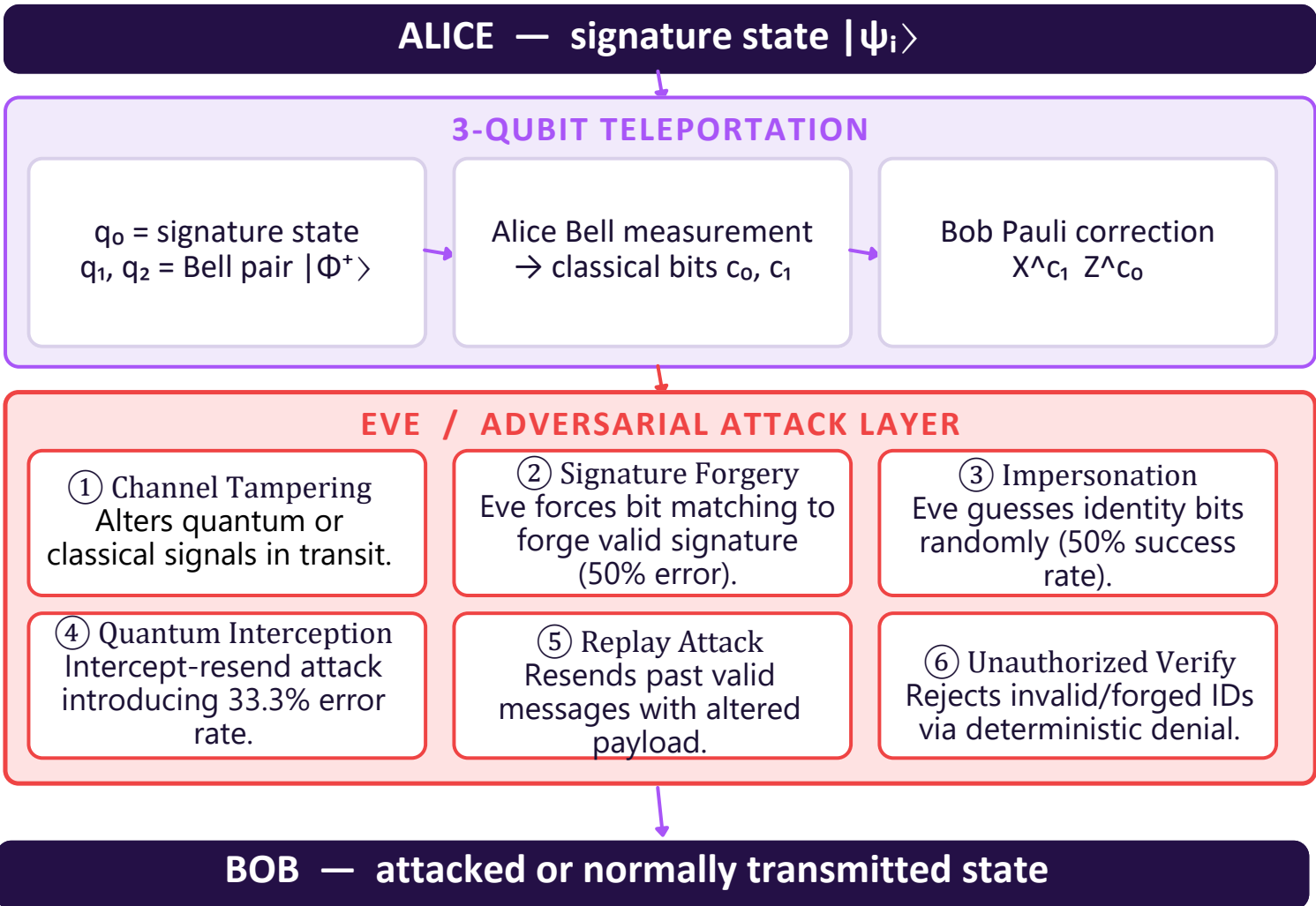
TECHNICAL INNOVATION / UNIQUENESS

- **Classification, not just detection:** basis-resolved signature (e_Z, e_X, e_Y) plus X-basis immunity and error-to-key correlation. 18/18 correct at n = 256.
- **Quantified security:** $P_{\text{forge}} = 5.64 \times 10^{-59} \rightarrow 193.5$ bits, decaying exponentially in n.
- **Defence in depth:** physical $\rightarrow$ quantum $\rightarrow$ statistical $\rightarrow$ protocol (nonce, HMAC authorization).
- **Zero AI / ML:** deterministic scoring against analytic signatures — every verdict is auditable.

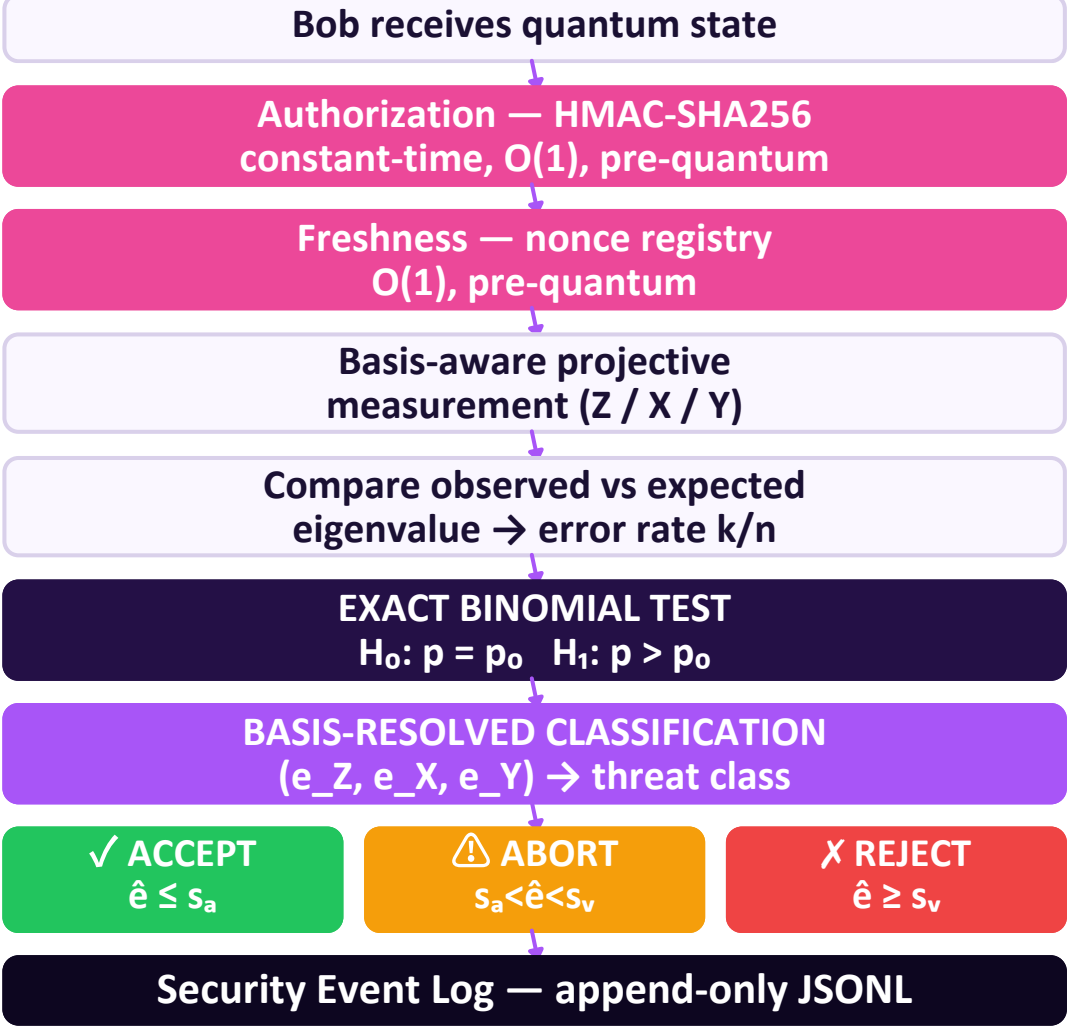
STAGE 1: SIGNATURE GENERATION



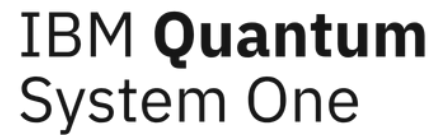
STAGE 2: QUANTUM CHANNEL & ATTACK LAYER



STAGE 3: VERIFICATION & DECISION



QUANTUM CORE



IBM Quantum QPUs — Heron
r2 (ibm_fez, ibm_marrakesh,
ibm_kingston, 156q)

HARDWARE



STATS & COMPUTE



FRONTEND



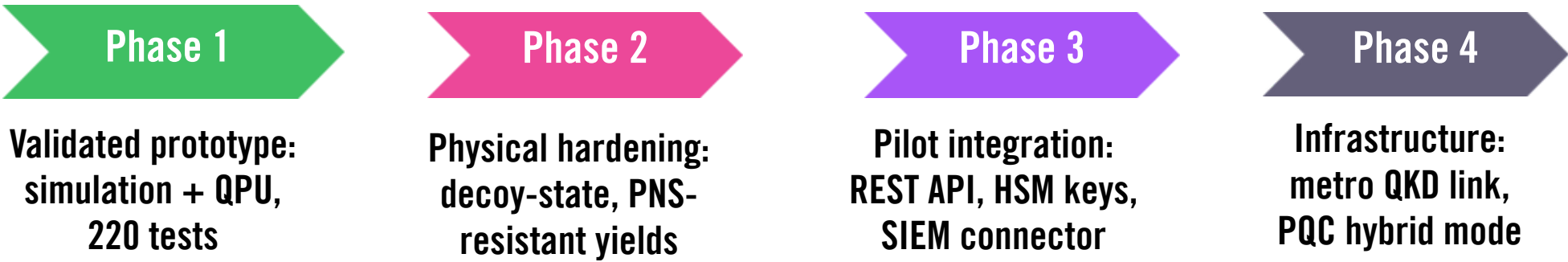
DOCTESTS

hashlib SHA-256
HMAC-SHA256
secrets CSPRNG

CRYPTOGRAPHY

Feasibility Analysis			
Theoretical		Empirical	
Deployed	1 Engineering Maturity 220 passing unit tests: analytic error rates, classification accuracy, forgery bounds against independently computed exact values, BBM92 QBER against theory	2 Functional Prototype Streamlit app across 12 sections – protocol, key distribution, quantum lab, hardware validation, security lab, classification, bounds, performance, audit.	
	4 Measured Efficiency Verification is $O(n)$ — confirmed empirically at exponent $k = 1.01$, $R^2 = 0.9997$, ≈ 1.7 ms per position.	3 Hybrid Execution Ideal Aer simulation, calibrated IBM noise models (offline, ~ 2 s), and live QPU submission.	

SCALABILITY & DEPLOYMENT ROADMAP



Quantum Security Risks and Mitigation Strategies		
Limited Runtime Restricted QPU runtime and queue availability.	→	Simulation and Validation Use local Aer noise simulation and real QPU for validation
Quantum Noise Presence of inherent quantum hardware noise	→	Noise-Aware Calibration Calibration and three-way rule to tolerate noise
False Results Potential for false positives or negatives.	→	Binomial Testing Exact Binomial testing with reported power and size
Limited Resources Constraints on available qubit resources.	→	Primitive Reuse Reuse of lightweight 3-qubit teleportation primitives
Variable Attacks Unpredictable nature of attack variability.	→	Parameterized Sweeps Execution of parameterized attack-strength sweeps
Replay Vulnerability Risk of replaying valid signatures.	→	Session Binding Session-nonce binding and verifier registry rejection
PNS Attacks Vulnerability to physical interception (PNS).	→	Decoy-State Protocol Implementation of decoy-state protocol at hardware layer
Scaling Challenges Difficulty in maintaining system scalability.	→	Modular Architecture Use of modular protocol and evaluation architecture

Enabling measurable, interpretable and quantum-ready verification of digital signatures

1

National & Critical Infrastructure

Evaluates quantum-secure signature verification across government, defence and critical digital infrastructure — with quantified forgery resistance (193.5 bits at $n = 256$) rather than assumed hardness, and an auditable evidence trail for every verdict.

2

Financial & Enterprise Security

Supports detection analysis for high-value transactions and enterprise authentication, where forgery, impersonation and replay carry direct financial consequence. Sub-2 ms per-position verification and $O(n)$ scaling make this operationally realistic.

3

Auditable & Regulator-Ready

Quantum measurement statistics and explicit mathematical decision rules instead of black-box AI/ML. Every decision reproducible from a logged seed and parameter set — directly serviceable for audit, forensics and regulatory review.

KEY BENEFITS

QUANTUM READINESS

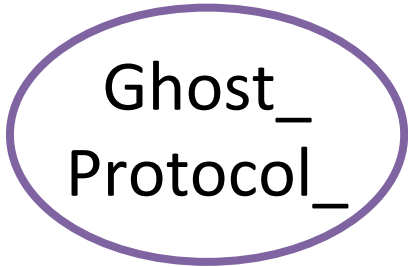
- Evaluates signature security against quantum-era threats
- Practical, runnable framework for teleportation-based QDS
- Enables experimentation ahead of large-scale quantum deployment
- Roadmap to physical-layer hardening (decoy-state / PNS)

SECURITY & TRUST

- Detects AND classifies forgery, impersonation, replay, channel manipulation, unauthorized verification
- Quantified metrics: error rate, p-value, detection power, forgery probability
- Mathematically interpretable, reproducible decisions
- Append-only audit log with enforced secret hygiene

ACCESSIBLE RESEARCH

- Extensive parameter sweeps via classical simulation — no QPU quota consumed
- Validated on real IBM Quantum hardware
- Modular architecture — new attacks and protocols plug in cleanly
- 220-test regression suite makes extension safe



RESEARCH AND REFERENCES



COMPETITIVE ADVANTAGE MATRIX

Criterion	Classical PKI / PQC	AI/ML Detection	OUR SOLUTION
Security basis	Computational hardness	Learned patterns	Physical law + exact statistics
Forgery bound	Asymptotic, assumed	None	Exact: 5.64×10^{-59} (193.5 bits)
Identifies attack type	No	Partial, opaque	Yes — 6 classes, 18/18
Explainability	High	Low — black box	Full — every verdict derivable
False-positive control	N/A	Empirical, drifts	Exact: 0.0348 at $\alpha = 0.05$
Infrastructure cost	Low	Medium (GPU)	Low — simulation-first
Replay resistance	Nonce-dependent	Not addressed	Nonce-bound registry, $O(1)$
Auditability	Good	Poor	Append-only, seed-reproducible

PROJECT RESOURCES

- Live Demo

<https://quantumsecurity.streamlit.app/>
- GitHub

<https://github.com/Naman-Vasudev/QKD>

RESEARCH PAPERS

QUANTUM SECURITY & PQC

- [1] NIST FIPS 204 — ML-DSA (2024)
- [2] NIST FIPS 205 — SLH-DSA (2024)

TELEPORTATION / QDS

- [3] Gottesman & Chuang, Quantum Digital Signatures, arXiv:quant-ph/0105032 (2001)
- [4] Zhao et al., Teleportation-based CV-QDS (2023)
- [5] Thornton et al., QDS over Insecure Channels (2018)

DECOY-STATE & FOUNDATIONAL QKD

- [6] Bennett & Brassard, BB84, Proc. IEEE ICCSSP (1984)
- [7] Ekert, E91, Phys. Rev. Lett. 67, 661 (1991)
- [8] Bennett, Brassard & Mermin, BBM92, PRL 68, 557 (1992)
- [9] Brassard, Lütkenhaus, Mor & Sanders, PRL 85, 1330 (2000) — defines the PNS attack
- [10] Hwang, PRL 91, 057901 (2003) — originates decoy-state
- [11] Lo, Ma & Chen, Decoy State QKD, PRL 94, 230504 (2005)
- [12] Ma, Qi, Zhao & Lo, Practical Decoy State, PRA 72, 012326 (2005)